

servers concurrently. Unlike a SAN, a NAS provides file-level storage semantics to multiple compute servers, appearing as a file server to the operating system running on the compute server. Internally, the NAS file server would access the physical storage at block level to access the actual data, while this is transparent to the OS on the compute server, which is exposed only to file-level operations on the NAS server.

Hybrids of SAN and NAS also exist. Since there is no file system concept for SANs, various file protection and access-control mechanisms need to be taken care of in the OS running on the compute server. In case of NAS, file protections and access control can be enforced at the NAS server. The next concept we need to understand in the storage domain is Scale-up Storage vs Scale-out Storage, which we will discuss next month.

Remembering Aaron Swartz

It has been almost two months since the death of Aaron Swartz. Most of us would have read about the enormous outpouring of grief caused by this tragic loss. Aaron Swartz was a programmer first and foremost, and the reason I wanted to mention him in our column was not just because he was a well-known activist who fought for the freedom of information on the Internet, but because he is a sterling example of what differentiates a great programmer from the run of the mill. He had an enormous enthusiasm for building software that solves challenging problems. He was involved in the development of the RSS format, wrote the Web.py framework, and was a technical architect of reddit.com, just to mention a few examples of his work. He had a great

passion for expanding and sharing his knowledge with all developers. Rest in peace, Aaron.

My 'must-read book' for this month

This month's must-read book suggestion comes from one of our readers, Aruna Rajan. She recommends the book 'Introduction to Information Retrieval' by Christopher D. Manning, Prabhakar Raghavan and Heinrich Schütze. This book focuses on various information retrieval techniques, including the most popular one of Web search engines. The book is available online at <http://nlp.stanford.edu/IR-book/html/htmledition/irbook.html>. Thank you, Aruna, for your suggestion.

If you have a favourite programming book/article that you think is a must-read for every programmer, please do send me a note with the book's name, and a short write-up on why you think it is useful, so I can mention it in the column. This would help many readers who want to improve their coding skills.

If you have any favourite programming puzzles that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming and here's wishing you the very best! 

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The author is an expert in systems software and is currently in a happy state between jobs. Her interests include compilers, multi-core technologies and software development tools. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group 'Computer Science Interview Training India' at <http://www.linkedin.com/groups?home=&gid=2339182>.

OSFY Magazine Attractions During 2013-14

MONTH	THEME	FEATURED LIST
March 2013	Virtualisation	Virtualisation Solution Providers
April 2013	Open source Databases	Certification & Training Solution Providers
May 2013	Network monitoring	Mobile Apps
June 2013	Open Source application development	Cloud
July 2013	Open Source on Windows	Web Hosting Providers
August 2013	Open Source Firewall and Network security	E-mail Service Providers
September 2013	Android Special	Gadgets
October 2013	Kernel Special	IT Consultancy
November 2013	Cloud Special	IT Hardware
December 2013	Linux & Open Source Powered Data Storage	Network Storage
January 2014	Open Source for Web development and deployment	Security
February 2014	Top 10 of Everything on Open Source	IT Infrastructure